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Lab 4 – UI & UX



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UC: Desenvolvimento de Videojogos

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Level 1

A new project was created and named “DPJ_Lab4”, the scene was renamed to “WhyUI”.

A new canvas was created, inside it, a new image.

The image was anchored to the middle of the screen.

The image was altered to the one previously chosen and converted to a sprite.

The resolution was changed from free aspect to Full HD.

We renamed the canvas to “Background”.

Inside the Background image we created a TMP (Text Mesh Pro) and it is named “Title”.

A new canvas was created and named “QuizCanvas”, a new TMP was also created with a placeholder text.

Inside QuizCanvas, we created a button gameobject.

A new empty game object was created and named “Button Group” inside the Quiz Canvas, the vertical layout group component was added.

The previously created button was added to the game object, three new copies were created.

The prefab of the button was created and placed inside its designated folder.

Level 2

The buttons previously created were replaced with the prefabs.

A dedicated folder was created to centralize future scripts.

A new script was created with the following:
Name: QuestionSO;

It’s a scriptable object;

Utilizes the “CreatAssetMenu” parameter with the appropriate name.

The start() and Update() methos were deleted.

A new variable was created named “question”, a string, private and observable in the inspector.

On top of the variable, we defined the parameter to “TextArea[2,5]”.

A new array was created:
[SerializeField] private string[] Answer= new int[3];

A new private variable, observable in the Inspector, has been added as an array and given an appropriate name to represent the correct answer to a quiz question.

Level 3

A new scriptable object was created for a question and put inside the designated folder.

Inside the script “QuestionSO” some methods were created:
public string GetQuestion();

public int GetCorrectAnswerIndex();

public string GetAnswer();

The methods should be returned respectively:

- The variable of the question text
- The variable of the correct answer
- A specific element of the response array.

Inside the scriptable object, we went to the inspector and assigned the number of questions, the answers and the correct answer.

A new script was created named “Questionaire”.

The script was associated to the “QuizCanvas”.

The script has the following specifications:

- A private variable of type “QuestionSO”, observable in the Inspector;
- A private variable of type “TextMeshProUGUI”, observable in the Inspector;
- A private variable of type “int”, corresponding to the answer index

In the “Start()” method, we used a code to get the question text from the

“QuestionSO” variable.

In the “Questionnaire” component of “QuizCanvas”, the question text was associated with the Scriptable Object;

Level 4

A new private variable was added, of type GameObject, which is an array, where the

response buttons are contained.

In the Inspector, slots were created in the “Questionnaire” component of the QuizCanvas where the answer buttons created in previous levels are associated.

Inside the “Awake()” method, we rewrite the “TextMeshProUGUI” variable and assigned the value of the question text that was stored in the scriptable object.
Inside the “Start()” method,

Inside the script “Questionaire”, a new method was created with the following specifications:

- Has no return;
- Its name is “OnAnswerSelected”;
- Must have the int argument named “Index”;
- Must have a validation that verifies if the argument value of the function is equivalent to the correct answer that is stored in the “QuestionSO” variable, If yes, the text to be displayed in the game should be changed to something that indicates whether the choice was correct—for example, “Correct!”;

To each of the previously created buttons, the permission to use the respective “OnClick()” was granted, the “QuizCanvas” gameobject was associated, the right dropdown that was chosen was the method “OnAnswerSelected” from the “Questionaire” script and the correct index value was assigned.

The verification block was expanded in the “Questionnaire” script by adding an “else” statement, which should follow these specifications:

- The variable corresponding to the index of the correct answer was assigned to the value of the correct answer from the ScriptableObject, retrieved via the GetCorrectAnswerIndex() method;
- A local variable named “correctAnswer” was created and assigned the value returned by the ScriptableObject’s GetAnswer() method, using the previously created variable as an argument;
- The question text was updated to a message indicating that the chosen answer is incorrect—for example, “Sorry, but the correct answer is: ”—followed by the value stored in the before mentioned local variable.

A new Void Method named “DisplayQuestion()”, the start content was moved to the new method.

The “DisplayQuestion()” method is called in the “Start()” method.

A new method was created with the following specifications:

- Its name is “SetButtonState”;
- It is a method that has no return type;
- Its argument is of type bool and is named “state”;
- It contains a “for” loop that iterates, incrementally, through the array of response buttons;
- Within the loop, each button is stored in a local variable of type Button, retrieved using the GetComponent method;
- Still within the loop, the button has its interaction state updated to match the value defined by the argument—that is, it should be set to “state”;

At the end of the OnAnswerSelected() method, the SetButtonState method is called with the value “false” passed as its parameter;

Level 5

In the “Questionnaire” script, a void method was created named “GetNextQuestion()” that calls the “SetButtonState(true)” method and the “DisplayQuestion()” method.

In the “Start()” method, “GetNextQuestion()” method is called.

Within the “QuizCanvas,” an “Image” Game Object was created and properly placed and it is a circle.

Its “Image Type” was changed to “Filled.”

“Fill Origin” was set to “Top.”

A new child Game Object was created under “QuizCanvas” named “Timer,” and a new script was attached, also named “Timer”, to it.

We defined a float variable named “timerValue,” another named “timeToCompleteQuestion,” and a third named “timeToShowCorrectAnswer.”

A method with no return type (void) was created named “UpdateTimer().” This method decrements “timerValue” (using “Time.deltaTime”) and checks if “timerValue” has reached or fallen below zero. If it has, resets “timerValue” to the duration allowed before the question becomes inaccessible.

In the “Update()” method, “UpdateTimer()” method was called.

A public boolean variable was created named “isAnsweringQuestion,” with a default value of “false.”
The “UpdateTimer()” method was modified and now it includes a check that validates the value of “isAnsweringQuestion”

The “if” statement validates the current time remaining to answer.

Before the time allotted for displaying the correct answer begins, a line of code was added that sets “isAnsweringQuestion” to “false”.

For the case where “isAnsweringQuestion” is “false”, copy the previously written code, but change the value of “isAnsweringQuestion” to “true”.

The first line, an if statement was created, it verifies if the variable "IsAnsweringQuestion" is true,

Inside it, the "timerValue" was decreased using the method "Time.deltaTime",

A debug.log was added to verify if the time is decreasing.

Another if statement was added; it verifies if the "timerValue" is equal or below zero.

Inside it the value of the variable "isAnsweringQuestion" is set to "false".

"timerValue" is set to equal the value of the "timeToShowCorrectAnswer".

An else statement is created for the if statement that was created on the start,

Inside the else statement, all the code created previously created is repeated with the exception of the "isAnsweringQuestion" that is set to true and "timerValue" is set to equal the value of the "timeToCompleteQuestion".

A new "float" variable was created named "fillFraction".

Inside the "if" block that validates "isAnsweringQuestion", the following line of code was added:

"fillFraction = timerValue / timeToCompleteQuestion;"

For the "else" case of that same block, the following line of code was added:

fillFraction = timerValue / timeToShowAnswer;

A new public bool variable was created named "loadNextQuestion";

If "isAnsweringQuestion" is false, once it becomes true and "timerValue" is updated, "loadNextQuestion" is also set to true;

Another public method was created called "CancelTimer()", which sets the timer value to zero, effectively performing a reset;

In the "Questionaire" script, variable of type "Timer" was created and its named "timer" and a variable of type "Image" named "timerImage".

In the "Start()" method, an object of type "Timer" was retrieved from the Unity scene using the "FindObjectOfType<>()" method;

In the "Update()" method, The "fillAmount" property of "timerImage" was linked to the "fillFraction" value of the "timer" variable;

Still within the Update() method, below the line created previously, an “if” block was added that checks the “loadNextQuestion” state of the “timer” variable. If this condition evaluates to true, the GetNextQuestion() method is called, and the “loadNextQuestion” value is set to false.

The CancelTimer() method was called at the end of the OnAnswerSelected() method.

In the “Questionaire” script, a boolean variable was created and its named “HasAnsweredEarly”.

In the “Timer” script, the access modifier of the isAnsweringQuestion boolean was changed to public.

In the Update() method of the “Questionaire” script, the “if” block was extended by adding an “else if” block to perform a secondary check: verify whether “HasAnsweredEarly” is true and “isAnsweringQuestion” is also true.

When both conditions are met, DisplayAnswer(int index) method is called.

Inside this “if” block, the following line of code was added at the very beginning:

“hasAnsweredEarly = false;”

The “DisplayAnswer” method contains the “if-else” logic previously found in “OnAnswerSelected;” it is called from there to replace the code that was originally in that location.

Still within the “else if” block, the “SetButtonState()” method was called, passing false as its argument.

In the OnAnswerSelected() method, thefollowing line of code was added at the beginning:

“hasAnsweredEarly = true;”